

RF-Sign: Position-Independent Sign Language Recognition Using Passive RFID Tags

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Abstract—Nowadays, sign language is becoming increasingly important in people's daily life. Existing solutions are often based on wireless signals (e.g., acoustic, visible, and WiFi) or wearable sensors to recognize gestures, but they suffer from vulnerability to environmental influences, poor security, and high energy consumption, which prevent them from accurately capturing finger micromovements. In this article, we propose RF-Sign, which uses passive radio-frequency identification (RFID) tags to capture multiple finger micromovements simultaneously to enable sign language support. In particular, two main issues are studied. One is the problem of positional differences when users make the same gesture, and the other is the problem of segmenting consecutive gestures using only empirical thresholding methods and ignoring the existence of differences in thresholds for different gestures. For position differences, we propose position models to normalize the hand's horizontal rotation angle and radial distance. For segmenting consecutive gestures, we use the received signal strength (RSS) trend of the reference tag to represent the finger micromovements state. The experimental results show that the average accuracy reaches 92.81% under different angles, distances, and other conditions.

Index Terms—Finger micromovements, radio-frequency identification (RFID), received signal strength (RSS), sign language.

I. INTRODUCTION

CURRENTLY, sign language is gaining more and more significance in people's daily lives. It is gradually transitioning from capturing coarse-grained hand movements to fine-grained finger movements, enabling gesture-based non-contact human-computer interaction (HCI). Additionally, sign

language is used in combination with facial expressions to alleviate communication barriers between individuals with hearing impairment and others.

Traditional hand gesture recognition systems [1], [2], [3], [4], [5], [6] use computer vision (CV) technology with cameras to detect hand shapes. However, they have limitations as they rely on light and might raise privacy concerns. Other methods [7], [8], [9], [10], [11] use gyroscope or accelerometer data to track hand movements, but these can use a lot of energy.

To address these issues, prior research has focused on wireless signal-based gesture recognition technology [12], [13], [14], [15], which is the central theme of this article. These approaches typically require users to perform hand movements, identifying finger micromovements as interactions for HCI. Generally, these wireless signal-based methods recognize gestures by observing how specific gestures affect the wireless channel, causing changes in RF signals over time, including amplitude, phase, and Doppler shift. Utilizing extracted features [16], [17], [18], these systems can identify gestures using predefined similarity measures such as dynamic time warping (DTW) based on template matching. However, such approaches are time intensive and computationally demanding. Additionally, certain methods employ distance-based classifiers with one or two classes of signal features to distinguish finger micromovements. Yet, these techniques rarely account for the impact of the distance and angle between the antenna plane and the hand, which can potentially degrade classifier performance.

Recent research has demonstrated that radio-frequency identification (RFID) technology can offer both device-free sensing [15], [19], [20], [21], [22], [23] through multipath effects, and precise hand motion detection via device-based solutions [16], [17], [24], [25]. This article introduces RF-Sign,¹ a passive RFID tag-based device system designed to capture RF signal changes during various finger micromovements. In contrast to prior studies that primarily focused on individual gesture recognition or roughly tracking the human torso, RF-Sign emphasizes continuous gesture recognition. This involves a more robust approach to continuous segmentation and fine-grained finger micromovements. As depicted in Fig. 1, RF-Sign detects finger micromovements through RFID tags worn on users' hands, which are then translated into understandable language and displayed on the screen. Consequently,

¹The source code and dataset for RF-Sign are provided at the following link: <https://github.com/ht0102/RF-Sign.git>.

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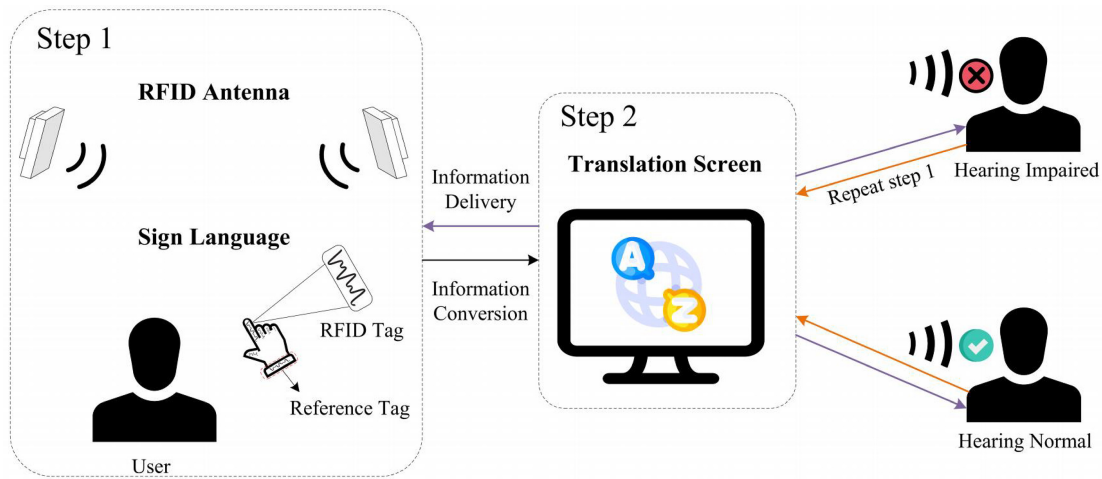


Fig. 1. Typical application of RF-Sign. According to step 1, the user performs a finger micromovement. Step 2 is to transfer the captured information to the translation screen for presentation to the hearing impaired or others. They can also repeat step 1 to communicate effectively with the user.

this significantly enhances communication efficiency between individuals with hearing impairments and others.

The specific idea is that RF-Sign recognizes hand gestures by continuously measuring signal changes during micromovements of the finger through RFID devices. Two significant challenges have been recognized.

The first challenge is that it is difficult to segment continuous gestures using only empirical thresholds. This is because different finger micromovements lead to different phase changes, and then the threshold should conform to the dynamic changes. If the threshold value set during gesture segmentation is too small or too large, then the starting and ending points of the gesture cannot be detected accurately. To solve this challenge, we set a reference tag at the wrist, determine the finger micromovement state by the received signal strength (RSS) trend of the signal from the reference tag, and record the micromovement state change point. Therefore, we can use the RSS trend of reference tag to segment different gestures, and then use the phase change to construct a phase matrix for sign language recognition.

The second challenge involves addressing the fluctuations in the phase profile across different distance angles. Our observations revealed that the rotation angle along the horizontal axis (parallel to the antenna pair plane) and the radial distance (perpendicular to the antenna pair plane) of the hand exhibited varying degrees of change in relation to the phase profile, even when the user executed identical finger micromovements. Consequently, attempting to directly recognize continuous gestures using the initial phase profiles results in a reduction in classifier performance. To address this challenge, a position-invariant phase matrix reconstruction solution is proposed. The approach introduces a location model to describe the relationship between hand phase, rotation angles, and radial distances. Specifically, hand rotation angles are estimated based on the phase difference between two antennas. Using the hand's rotation angles, a specified movement is performed outside the beam from the antennas to the center to find the radial distance of the hand. Therefore, the location model enables the accurate reconstruction of the

phase matrix, precisely describing the phase variation of finger micromovements.

RF-Sign is implemented using a universal software radio peripheral (USRP) device and passive tags. Comprehensive performance evaluations were conducted across various positions, encompassing different rotation angles and radial distances. RF-Sign attains an average angle measurement error of 4.3° and a distance measurement error of 3.24 cm. Additionally, the average gesture recognition accuracy surpasses 92.81% for both metrics.

The main contributions are summarized as follows.

- 1) A novel gesture segmentation method is proposed to dynamically segment gestures using the reference tag RSS changes to determine the finger micromovements states. In addition, the reference tag eliminates the effect of arm micromovements when the user performs a gesture.
- 2) A position model estimating the horizontal hand rotation angle and radial distance to reconstruct the finger micromovements phase matrix is proposed to improve the flexibility of RF-Sign by analyzing the effect of different hand positions on the phase.
- 3) We have built a prototype RF-Sign using USRP to help hearing-impaired people communicate with others. Extensive experiments show that RF-Sign enables accurate and robust gesture recognition with an average accuracy of 92.81%.

II. PRELIMINARIES

In this section, we introduce the RFID basics and the guidelines for constructing RFID sensing planes.

A. RF Phase and RSS

In RFID communication, the RF signal travels back and forth between the antenna and the tag, passing a total distance of $2R$. Assuming an RF carrier wave of frequency $f(\text{Hz})$, then the relationship between frequency and wavelength can be seen by $\lambda = (c/f)$, where c is the electromagnetic

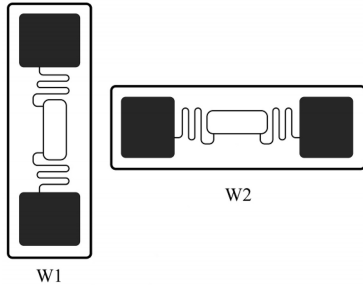


Fig. 2. Reference tag deployment.

wave speed, and in the air environment, c is the light speed ($\approx 3 \times 10^8$ m/s). In addition to the phase variation with distance, the reader's transmission line and the tag's reflection characteristics introduce additional phase errors θ_R and θ_{TAG} , respectively. Then, the relationship between phase Φ and distance can be expressed as

$$\Phi = \left(2\pi \frac{2R}{\lambda} + \theta_R + \theta_{TAG} \right) \text{MOD } 2\pi \quad (1)$$

where MOD 2π is used to ensure the periodicity of the measured phase.

The RSS of the reflected signal cannot be neglected and the effect of the signal power P (mW) is usually considered. The RSS can be expressed as

$$\text{RSS} = 10 \lg P = 10 \lg \left(\frac{P}{1 \text{ mW}} \right) \quad (2)$$

where RSS is generally negative and maximum 0. In the ideal case, the reader transmits the signal and the receiver receives it intact.

B. Tag Selection and Deployment

In implementing gesture recognition, a pivotal challenge revolves around configuring tags to capture finger micromovements. As a practical approach, we affix the tags to our fingers. Upon executing a gesture, each tag initiates a signal alteration, indicative of the movement performed.

1) *Tag Selection*: There are a variety of commercial tags designed with different sizes and attributes. Among them, we choose NXP-7018 tags. The NXP-7018 size (73 mm $\times$ 21 mm) conforms to the finger geometry and has sufficient recognition range (≈ 12 m). Therefore, the NXP-7018 is ideal for capturing finger micromovements.

2) *Reference Tag Deployment*: In the introduction section, we introduce the concept of the reference tag's position. In this section, we proceed with comparison experiments aimed at validating the appropriateness of the reference tag's placement.

Preliminary Experiment: USRP can report the channel parameters, i.e., RSS and phase, for each tag being queried. To explore the impact of the reference tag's deployment position on the signal, a series of controlled experiments were carried out. Volunteers were engaged to perform dynamic gestures, and subsequently, RSS and phase data were collected for various tag positions.

Sensing Ability Study: The capability to sense finger micromovements is being tested. Figs. 3 and 4 compare the phase and RSS for different tag locations (specifically, W1 and W2 as

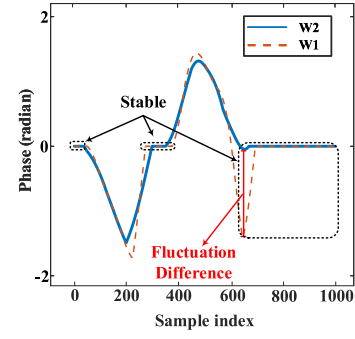


Fig. 3. Impact of reference tag placement on phase.

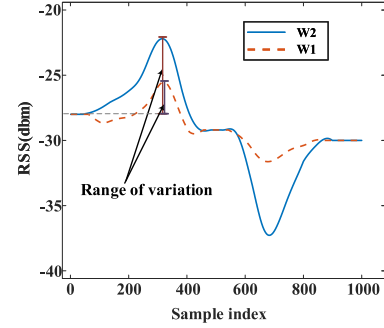


Fig. 4. Impact of reference tag placement on RSSI.

shown in Fig. 2). It is noticeable that the phase of W1 exhibits more significant fluctuations than that of W2 during a steady phase state. This observation underscores the higher stability of W2. Moreover, W2 stands out when the RSS changes, indicating that it maximizes the reception of the reader's transmit signal and performs well in detecting environmental dynamics.

III. SYSTEM DESIGN

The system encompasses several main components, as illustrated in Fig. 5. The key components include signal preprocessing modules, with the core ones involving phase profile construction, continuous gesture segmentation, and sign language recognition. In detail, upon receiving the time series signal from each marker finger, the received signal undergoes interpolation and filtering to enhance smoothness. Subsequently, the finger micromovements are transformed into a phase matrix format. However, variations in hand position during similar finger micromovements might degrade recognition performance. Therefore, the gesture is represented through constructing a position-independent phase matrix. Additionally, the changing trend of the reference tag's RSS is employed to segment continuous gestures, and the resulting segmented phase matrix serves as a template. To ensure template accuracy, matrices with outliers are eliminated empirically, and representative templates are chosen using a voting scoring approach. Ultimately, RF-Sign utilizes a bidirectional long short-term memory (Bi-LSTM) model to identify representative templates.

A. Signal Preprocessing

Interpolation: Due to the hand moving too fast or the antenna pitch is not reasonable (if the pitch is small, the

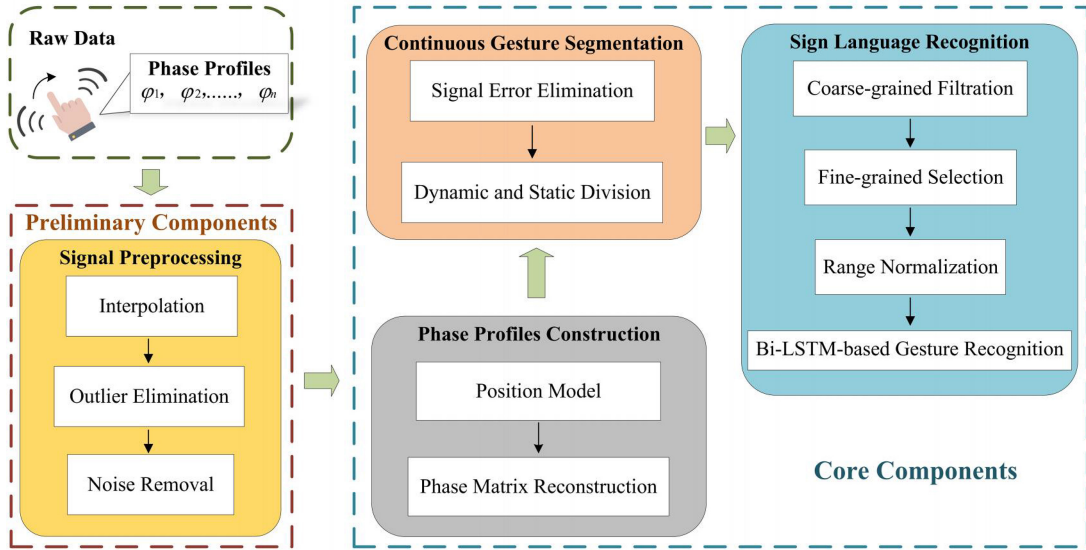


Fig. 5. System architecture.

transmitted signal is directly received by the antenna, whereas if the pitch is large, the tag may be in the blind spot of the antenna to the center beam), the tag cannot reflect the signal in time, which brings the problem of missing measurement values. We need to define three variables, first determine the interval number t , the interval to ensure the presence of missing values, must meet $t_k \leq t < t_{k+1}$ ($k = 1, 2, 3, \dots, n$), where t_k and t_{k+1} denote breakpoints. In order to make the local monotonicity constant, then determine a local variable λ and the mean difference δ_k , which is calculated as

$$\lambda = t - t_k \quad (3)$$

$$\delta_k = \frac{\theta_{k+1} - \theta_k}{t_{k+1} - t_k} \quad (4)$$

where θ_{k+1} and θ_k indicate the phase values corresponding to the current interval point and the next interval point. Finally, the interpolation function is defined as

$$F(\theta) = \theta_k + \frac{\theta_{k+1} - \theta_k}{t_{k+1} - t_k}(t - t_k) = \theta_k + \lambda\delta_k. \quad (5)$$

Outlier Elimination: The presence of outliers within the phase profiles leads to a decline in recognition accuracy. To address this, the Hampel filter is employed to assess data validity. This involves computing the absolute deviation from the median, utilizing both the median and standard deviation within a designated window. If outliers are identified, they are substituted with the median value from neighboring windows.

Noise Removal: High-frequency noise has an impact on the phase fluctuations produced by finger micromovements. To mitigate this, the weighted moving average filter is utilized to eliminate noise by averaging data in segments.

Illustrated in Fig. 6, a comparison is made between the phase profile processing before and after the execution of a single-finger micromovement. It is evident that the absent phase values are supplemented and that outliers and high-frequency noise are effectively eliminated through filtering.

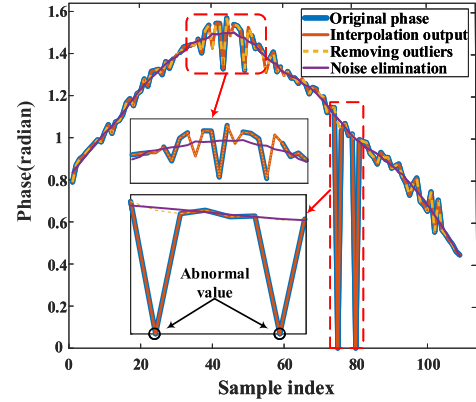


Fig. 6. Signal interpolation and smoothing.

B. Phase Profiles Construction

Factors unrelated to the environment, like the angle of hand rotation and radial distance (distance to the antenna pair plane), have the potential to impact system performance detrimentally. To address this, we introduce a position model that captures the correlation between phase and tag position, enabling the normalization of tag positions. Furthermore, we employ the phase equation derived from the position model to reconstruct the phase matrix.

1) **Position Model:** When executing identical finger micro-movements, we identified two distinct issues: 1) *Nonuniform Distance Values:* Theoretically, the hand's distance from the antenna should exceed half a wavelength (i.e., 16.4 cm), causing a phase change surpassing one full cycle, thereby leading to substantial phase errors and 2) *Undetermined Angles:* Executing the same gesture may result in varying rotation angles within the same plane, influencing the phase alteration, as depicted in Fig. 7.

In the context of intricate finger micromovements, the precision of gesture recognition can be impacted by deviations in both angle and distance. Angle variations dictate

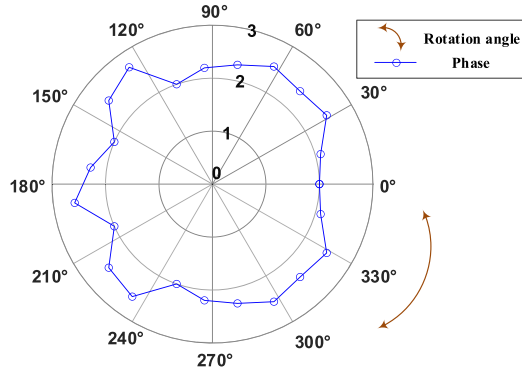


Fig. 7. Phase versus tag rotation angle: phase value is related to the tag rotation angle.

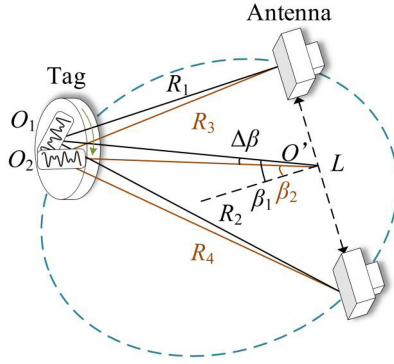


Fig. 8. Tag rotation causes the angle between the antenna and the tag to change.

the hand's horizontal displacement, while distance fluctuations pertain to vertical shifts. This implies that angle alteration has a direct impact on distance and, consequently, an indirect influence on phase. Thus, even when performing identical finger micromovements, phase alterations might differ across diverse positions. Hence, the normalization of tag positions is imperative for accurate finger micromovement recognition.

The proposed position model serves to illustrate the phase's fluctuation concerning hand angle and distance. Given that distance estimation necessitates knowledge of the angle between the tag and the antenna pair plane, our initial step involves angle estimation. As depicted in Fig. 8, angle discrepancies might arise during the user's gesture execution. Specifically, with the antennas positioned at a distance of L , the antenna pair's center at O' , the initial tag's center at O_1 , and the center point post angular deflection at O_2 . When the tag is in its starting position, the distances from the antenna's phase center are R_1 and R_2 , respectively. Assuming $L \ll R_1, R_2$, based on Fig. 8, it becomes apparent that $\sin \beta_1$ gradually diminishes as β_1 transitions from 90 to 0. This leads to

$$R_1 - R_2 \approx L \sin \beta_1 \quad (6)$$

specifying that β_1 is a negative angle when the tag is rotated along clockwise. Since the phase difference is used to calculate the angle of the tag, two antennas are needed to receive the reflected signal separately, so the expression of the distance

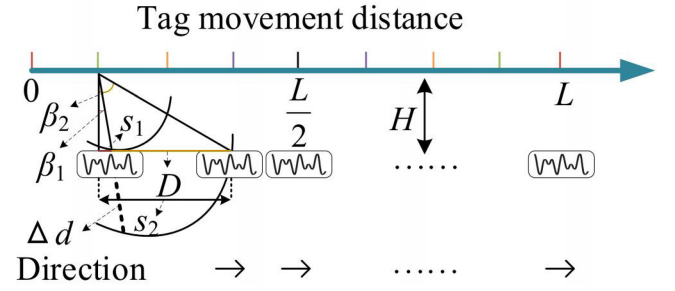


Fig. 9. Tag is moved to the specified position, i.e., four pairs of symmetric points, to obtain the distance from the tag to the antenna.

difference is

$$R_1 - R_2 = \frac{\lambda(\Delta\phi - \Delta\phi_0)}{4\pi} \quad (7)$$

where $\Delta\phi$ is the phase difference between the two antennas, and $\Delta\phi_0$ is the phase error. According to (6) and (7), we can get

$$L \sin \beta_1 = \frac{\lambda(\Delta\phi - \Delta\phi_0)}{4\pi}. \quad (8)$$

When L decreases, $\Delta\phi$ also decreases, but it may cause the antenna to fail to respond to the tag. To prevent $\Delta\phi$ from exceeding one cycle, then L satisfy that

$$L \leq \frac{\lambda}{2 \sin \beta_1} \quad (9)$$

where L takes a great value at β_1 is $(\pi/2)$ and does not exceed half of the wavelength.

Then, according to (8) and (9), it can be written as

$$\beta_1 = \arcsin \left[\frac{\lambda}{2\pi L} (\Delta\phi - \Delta\phi_0) \right]. \quad (10)$$

Similarly, we can find the angle β_2 after tag deflection, so the angle difference

$$\Delta\beta = \begin{cases} |\beta_1 - \beta_2|, & \text{Dir}(\beta_1) = \text{Dir}(\beta_2) \\ |\beta_1 + \beta_2|, & \text{Dir}(\beta_1) \neq \text{Dir}(\beta_2) \end{cases} \quad (11)$$

we define $\text{Dir}()$ as a function to determine the current corner direction as well as $|\cdot|$ is the absolute value function. Therefore, the current angle is recorded each time a finger micromovements is executed. Then the angle difference is calculated when the same type of finger micromovements is completed, thus ensuring the stability of the phase change.

The phase is directly influenced by the radial distance's magnitude, prompting the necessity for distance estimation. Specifically, the antennas' spacing, denoted as L , is partitioned into eight equidistant sections using the midline at $(L/2)$. Initially, the hand bearing the marker is positioned according to specifications, with the distance to the antenna being gauged at four equidistant points on each side, as illustrated in Fig. 9. Subsequently, a set $K = \{k_1, k_2, \dots, k_8\}$ of feasible values k is ascertained. To establish criteria, symmetry points in pairs exist on either side of the boundary line. The reasonableness of the distance can be affirmed if the distance error μ of these pairs adheres to $\mu \leq 10^{-4}$. It is worth noting that the radial distance, denoted as H , facilitates tag movement within a fixed

plane. Given the known values of radial distance H and arrival angle β , determining the phase change further requires computation of the tag's displacement D within the same plane, achieved using the formula $D = |s_1 - s_2|$, where s_1 and s_2 represent the distances from the current position to the final position and are computed as follows:

$$s_1 = H|\tan \beta_1|, s_2 = H|\tan \beta_2| \quad (12)$$

it can be seen that both s_1 and s_2 are linearly related to H . With the above equation, we can obtain the relationship equation between radial distance and moving distance

$$D = H||\tan \beta_1| - |\tan \beta_2|| \quad (13)$$

where β_1 and β_2 are in the maximum sector 2β corresponding to the reader, satisfying $\beta_1, \beta_2 \in [-\beta, \beta]$.

2) *Phase Matrix Reconstruction*: For each completed finger micromovements, the reader receives data about time t . First, we construct the phase profile of the five fingers as an initialized phase matrix

$$M = \begin{bmatrix} \theta_{11} & \cdots & \theta_{51} \\ \vdots & \ddots & \vdots \\ \theta_{1t} & \cdots & \theta_{5t} \end{bmatrix}_{t \times 5} \quad (14)$$

where θ_{ij} denote the phase value of a particular finger at a given moment.

The position model's representation of the phase-hand position interplay (angle and distance) necessitates accounting for the impact of hand angle and distance during finger micromovement execution. Subsequently, as illustrated in Fig. 9, we initiate the process by utilizing the initial phase ϕ . From there, we compute the phase difference $\Delta\phi$ between the current time point and the subsequent one. This computation enables us to formulate the equations that correlate $\Delta\phi$ with the arrival angle β , the moving distance D , and the radial distance H

$$d_2 = \sqrt{s_2^2 + H^2} = \sqrt{H^2(\tan^2 \beta_2 + 1)} \quad (15)$$

where d_2 signifies the Euclidean distance from the tag's center to the reader's center. Similarly, we can get

$$d_1 = \sqrt{(D + H|\tan \beta_2|)^2 + H^2}. \quad (16)$$

It is important to calculate the phase change $\Delta\phi = (4\pi\Delta d/\lambda)$ for the marker point's shift from the current position to the previous one, with $\Delta d = d_1 - d_2$, signifying the positive movement along the x -axis. Given the familiarity with the initial phase matrix M , we proceed to reconstruct the phase matrix as follows:

$$M' = \begin{bmatrix} \theta_{11} & \cdots & \theta_{51} \\ \vdots & \ddots & \vdots \\ \Delta\phi_{1,n} & \cdots & \Delta\phi_{5,n} \end{bmatrix}_{n \times 5} \quad (17)$$

where $\Delta\phi_{i,j}$ represents the phase alteration of the i th finger after the time interval j . This fresh phase matrix is capable of capturing the delicate motions of all five fingers concurrently. Its phase profiles epitomize the phase variations of each finger, implying that every column in the matrix denotes the phase change of the corresponding finger when executing the gesture.

C. Continuous Gesture Segmentation

Given the lack of precision in the initial and final positions of continuous gestures, two factors merit consideration: 1) the influence of arm oscillations while the hand moves and 2) the differentiation between stationary and moving hand actions. Consequently, we leverage the RSS trajectory of the reference tag to partition continuous gestures and mitigate the impact of tremors.

1) *Signal Error Elimination*: Recognizing that the signal received by the antenna at the end of a gesture encompasses static and dynamic components is crucial. The dynamic component primarily consists of the reflected signal resulting from finger micromovements, while the remaining portion constitutes the static component, which includes phase inaccuracies originating from other parts of the body. The arm can be divided into two segments: the upper arm and the lower arm, with the latter being closer to the hand. Vibrations in the lower arm can lead to hand displacement. Therefore, the static component mainly emerges within the lower arm's range, prompting us to select the wrist as the reference tag location. Since each gesture carries its own set of errors, these inconsistencies accumulate and diminish the accuracy of dynamic segmentation. By choosing the wrist as the reference tag's location, we posit that the total signal received by the antenna is represented as $S = \sum_{l=1}^5 S_{l,d} + S_s$, where $S_{l,d}$ signifies the reflected signal arising from all finger micromovements, and S_s denotes the reflected signal generated by arm oscillations. Consequently, when a finger micromovement is executed, the phase error θ_s is calculated and subsequently eliminated from the overall phase, ensuring that the processed phase $\theta_{l,d}$ accurately reflects the finger micromovements' state.

2) *Dynamic and Static Division*: Gestures are distinguished through the phase matrix of finger micromovements. However, an issue arises in accurately identifying segmentation boundaries during the execution of consecutive gestures. This challenge emerges from the distinct phase alterations associated with various finger micromovements, rendering the use of empirical threshold techniques for continuous gesture segmentation less precise. As close proximity between finger tags can lead to signal interference, we introduce a reference tag for segmenting continuous gestures. In our methodology, we leverage the trend of the reference tag's RSS to capture static and dynamic changes within the gesture. Precisely, we partition the gesture execution process into two stages: initiation and stabilization. The phase matrix encapsulates phase shifts during finger motion. Given its central role in finger micromovements identification, we initiate phase profile alignment through a differential operation on the RSS. Subsequently, we establish a moving window, denoted as x , with a size of 10, and compute the variance value encompassed by the window. As depicted in Fig. 10, multiple paired local wave peaks manifest, and certain irregular wave peaks emerge near spikes due to environmental influences originating from multipath effects. After multiple tests and adjustments, it has been determined that using an empirical threshold value of $Y(Y \geq 0.47)$ is reasonable. If the variance value is less than Y , then the wave is

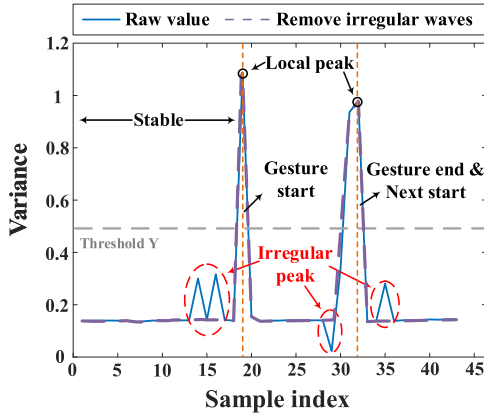


Fig. 10. Dynamic segmentation.

seen as an outlier to be eliminated, so that only pairs of spikes are retained.

Next, it is crucial to determine the beginning and end of a gesture. When the gesture is not started, the signal strength is at a plateau, once the new gesture is made, the peaks appear. Then, the current gesture remains unchanged so that the signal strength will level off again until the next new gesture arrives. That is, the position where the current gesture starts corresponds to the end of the smooth phase of the previous gesture. The positions of all starting points are recorded, and the phase matrices with different row widths are cut out by phasing the sequential gestures, as well as the matrices expressing the same gestures, are placed in a set $N_i = \{N_1, N_2, \dots, N_n\}$, where n denotes the n th gesture. In the robustness testing designed in Section IV, the RF-Sign system undergoes multiple adjustments and passes the tests, demonstrating that the segmentation algorithm for continuous gestures is suitable for the current scenario.

D. Sign Language Recognition

In order to better distinguish between sign languages, we aim to select a representative phase matrix, which is a key condition for the successful differentiation of sign languages.

1) *Coarse-Grained Filtration*: The phase matrix within a specific set N is examined to assess its feasibility, gauging the tag's ability to appropriately respond as the gesture is executed within the set N . Various factors can contribute to unresponsiveness, such as tag bending, which reduces the receiving area, or the tag's orientation opposing the antenna beam's direction, leading to nonactivation of the internal antenna.

2) *Fine-Grained Selection*: We have to consider the phases of specific fingers in different matrices, when the phase matrix in the set N is normal, but there are still phase profiles with different trends of variation, and the same kinds of sign languages tend to show phases with similar trends of variation. As shown in Algorithm 1, the length of the phase profile varies because the user is not guaranteed to execute the gesture at the same speed each time. First, set all phase profiles to the same starting point. Define variables for position records POS , total score sum , and initialize them to 0. Second, the tail complementary zero operation is performed. Then, we will

Algorithm 1 Select a Representative Phase Matrix

Require: Initialize all phase profiles in the phase matrix M' of the set N

Initialize the variable $POS = 0$

Initialize the total score $sum = 0$

Output: representative sequences

- 1: Set all phase profiles to the same start point
- 2: Tail complementary zero operation for phase profiles
- 3: Determine the upper boundary b_1 and the lower boundary b_2
- 4: Derive at each pair of boundaries $b_i = b_{i,1}, b_{i,2}$ and obtain the trend series γ
- 5: Set up voting scoring rule: compare sequences γ to the same position
- 6: **for** POS is less than the length of the sequence γ **do**
- 7: **if** γ_i and γ_{i+1} are not the same at POS **then**
- 8: sum plus 1 and POS plus 1
- 9: **else**
- 10: sum subtract 1 and POS plus 1
- 11: The total score sum of each sequence is recorded, and then the top G ones with high total scores are selected.
- 12: **return** representative sequences

determine the upper boundary b_1 and the lower boundary b_2 , and then derive the derivative at the boundary to obtain the sequence of change trends $\gamma = \{\gamma_1, \gamma_2, \dots, \gamma_n\}$, γ_i is divided into three categories: 1) increasing (+); 2) no fluctuation (—); and 3) decreasing (—). Finally, we propose a voting scoring rule, which involves comparing sequences γ between the same positions POS . POS increases by 1 sequentially but does not exceed the length of the sequence. If the trend categories are the same, one point is added. Conversely, one point is subtracted. The total score sum of each sequence is recorded, and then the top G ($\lfloor n/4 \rfloor \leq G \leq \lfloor n/2 \rfloor$) with the highest scores is selected.

3) *Range Normalization*: If the distance between different phase profiles is too large, then it may lead to incorrect classification results. We set each phase profile θ_t , where t denotes time and varies in size. Then, the phase profiles are normalized for distance to the range $[0, 1]$

$$\hat{L} = \frac{L - \min(L)}{\max(L) - \min(L)} \quad (18)$$

where $\hat{L}$ and L are the updated and original time series, respectively, and the $\min()$ and $\max()$ functions are the minimum and maximum values in L . Thus, the phase matrix after the above operation is able to accurately represent its corresponding gesture.

4) *Bi-LSTM-Based Gesture Recognition*: We use the phase matrix of multiple sets N as the feature representation of sign language. There are two reasons why we chose Bi-LSTM. On the one hand, the recognition accuracy can be improved because Bi-LSTM can handle the spatiotemporal features of finger micromovements while considering the speed of hand movement and the user's height. On the other hand, Bi-LSTM does not require a large enough data set to support the

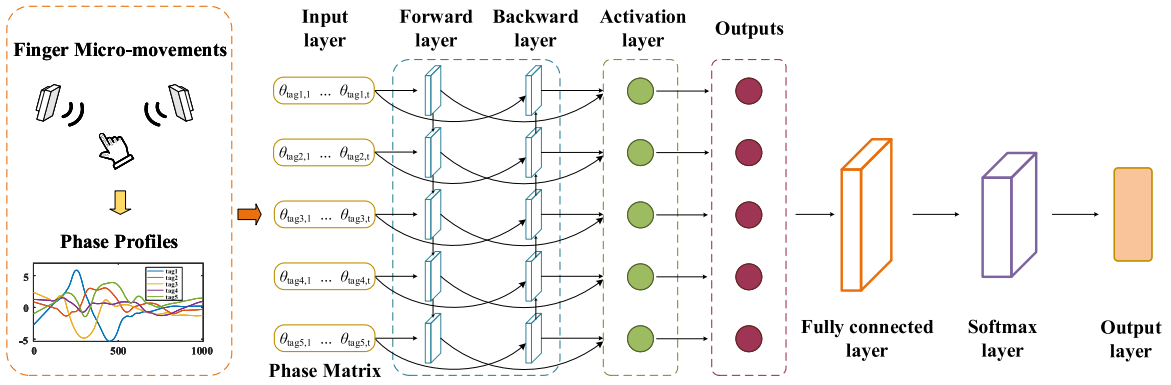


Fig. 11. Network architecture.

model's training due to its uncomplicated architecture. And the computation time cost is low. The basic architecture of the classification network within the RF-Sign system is illustrated in Fig. 11. The network takes a 5-D phase matrix as input, and the final scores for each finger component contribute to calculating the overall classification score for each gesture. The classification model comprises three hidden layers, encompassing the forward layer, backward layer, activation layer, fully connected layer, and softmax layer. Notably, the forward and backward layers are pivotal components, both leveraging the LSTM model's memory-retaining capabilities by propagating each phase matrix twice. This establishes a strong correlation between the phase characteristics of each finger, thereby ensuring the precision of sign language classification. With its adeptness in learning classification tasks, the Bi-LSTM enables the automatic alignment of phase profiles of varying lengths. Consequently, accurate identification of finger micromovements can be achieved during the continuous segmentation of gestures, even when dealing with phase profiles of differing lengths.

IV. SYSTEM EVALUATION

In this section, we present our system implementation and evaluate its performance using several gestures and users.

A. Experimental Setup and Methodology

Experimental Setup: The experimental setup of the RF-Sign system consists of a dual-antenna USRP device and several wearable RFID tags, as shown in Fig. 12. USRP can simulate any signal through a signal designer, making it convenient for data synchronization and developing data analysis algorithms. The four-channel USRP device contains an internal X310 daughterboard and a pair of circularly polarized directional antennas with a gain of 6.5 dBi arranged in compliance with the EPC C1G2 standard. The distance between RFID antennas is 100 cm in order to query the scanning tags. The USRP operates in channel hopping mode on frequency band 902–920 MHz. The user wearing the RFID tag performs finger micromovements at 150 cm in front of the antenna pair. Specifically, RF-Sign measures the phase variations of finger micromovements, and to achieve this, a glove design is implemented. The glove is equipped with five 7018-U9 RFID wet

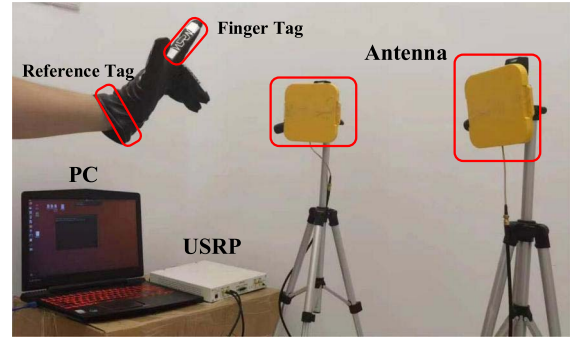


Fig. 12. Experimental deployment.

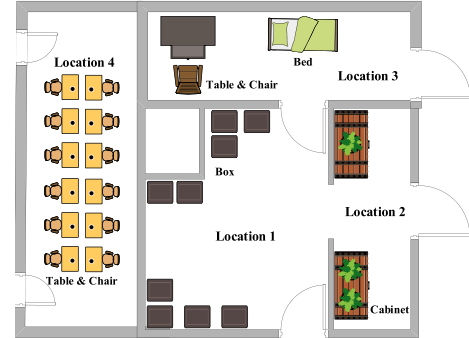


Fig. 13. Plan view of different locations.

inlay tags that are attached to the hand. These lightweight tags ensure that users do not experience a sense of weight while wearing the glove. An identical tag is attached to the wrist at a distance of 6 cm from them, mainly to detect environmental changes and abnormal arm shakes.

Experimental Methodology: To validate the effectiveness of RF-Sign in recognizing finger micromovements, we design an experimental environment. As shown in Fig. 13, Location 1 is represented. The layout consists of walls surrounding an area measuring 5.32 m in length, 2.56 m in width, and 4.33 m in height. Several cardboard boxes measuring 1 m in length, 0.5 m in width, and 0.7 m in height are placed in the corners. A total of ten participants take part in the experiment, including seven males and three females. The age range of the male participants is primarily between 23 and 30 years old,

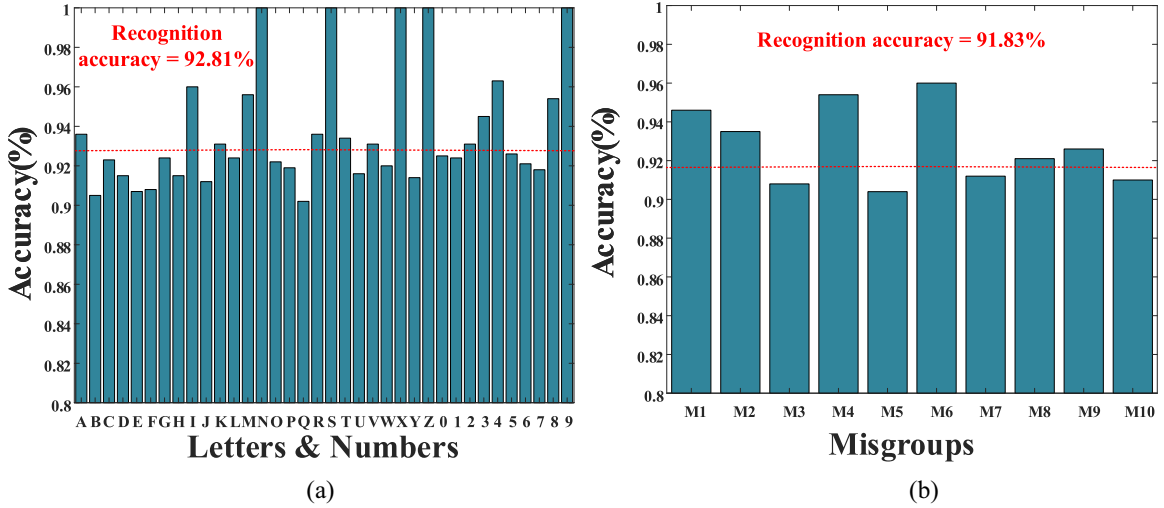


Fig. 14. Evaluation of micromovement recognition results. (a) Recognition accuracy of letters and numbers. (b) Recognition accuracy of misgroups.

with two additional participants aged 38 and 52 years old. The female participants are 27, 36, and 55 years old. This selection ensures representation across different age groups. It should be noted that male hands generally have larger dimensions compared to female hands. The distance from the wrist to the middle fingertip ranges from 18 to 24 cm for males, while for females, it ranges from 14 to 17 cm. Participants are instructed to don gloves equipped with RFID tags and a wristband before executing predetermined gestures. These gestures encompass the complete set of 26 alphabet letters, along with the Arabic numerals 0–9. Each participant is required to replicate all these gestures ten times consecutively.

B. Data Collection and Metrics

To evaluate the performance of RF-Sign in finger micromovements, we define four metrics.

Data Collection: To ensure the robustness of the RF-Sign system, data augmentation methods are used to supplement the data set. For hand movement speed, a resampling method is applied to each sign language, resulting in 21 600 new samples. In scenarios with obstacles in the environment, white noise is added using the appropriate method, generating 18 000 new samples. To address misclassified sign language groups, a combination of random zero-padding and moving average filtering is employed to emphasize the phase variation characteristics of each sign language, creating 28 800 new samples. As a result, a total of 72 000 samples are generated to ensure the adequacy of the data set.

Metrics: For sign languages represented by finger micromovements, the recognition accuracy is defined as (C_t/C_a) , where C_t and C_a are the number of correctly classified gestures and executed gestures, respectively. Precision, Recall, and F1 score definitions, respectively

$$\text{Precision} = \frac{TP}{TP + FP} \quad (19)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (20)$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (21)$$

where TP denotes true positive, TN denotes true negative, FP denotes false positive, and FN denotes false negative. In particular, we train a general Bi-LSTM model and then use this model to classify all the sign languages.

C. Micromovement Recognition Performance

RF-Sign shows good performance in recognition accuracy with an average accuracy of 92.81%. As shown in Fig. 14(a), the recognition rate of all letters is above 90%, the accuracy rate of Arabic numerals is above 92%, and the recognition accuracy rate of 21 out of 26 letters is above 91%. In particular, the misclassification rate is high because the gestures indicating the similarity between letters and Arabic numerals/letters can appear high. Therefore, other letters and numbers, due to their different presentation, can be recognized correctly 100% of the time.

To make RF-Sign more reliable, it is necessary to recognize the gestures with high similarity accurately. Specifically, we test data sets that may be misclassified. As shown in Table I, ten misgroups are set up, with Arabic numerals 0–9 corresponding to similar letters. As shown in Fig. 14(b), RF-Sign has an accuracy of over 91.83% in all misgroups, with a maximum accuracy of 96%. The table shows the following observations: for fine-grained finger micromovements, RF-Sign considers the position of the entire hand as well as environmental influences for micromovement recognition through phase changes.

The confusion matrix and other evaluation metrics are calculated to evaluate the RF-Sign system better. As shown in Fig. 15, the depth markings on the far right of the confusion matrix are to see the number of correct identifications for each sign language category. It can be seen that some of the sign language categories have relatively shallow depths due to the data set in the misgroups, using resampling and adding white noise to improve RF-Sign system performance. So after data set division, the number of misgroup validation sets is high

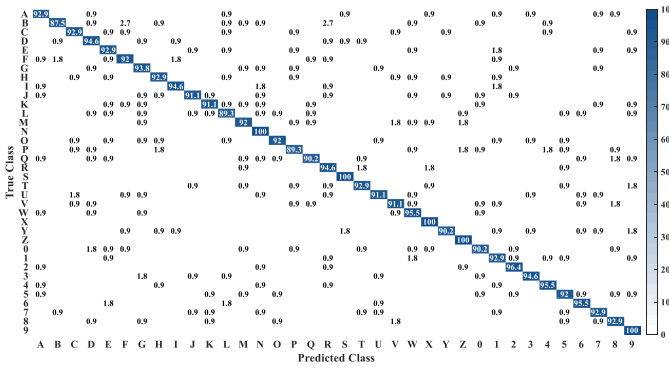


Fig. 15. Confusion matrix for letters and numbers.

TABLE I
MISGROUPS

Group	Numbers	Letters
M1	0	D O
M2	1	G I
M3	2	F H V
M4	3	E P
M5	4	B
M6	5	U
M7	6	Y
M8	7	Q
M9	8	L R
M10	9	J

TABLE II
PERFORMANCE METRICS OF RF-SIGN

Precision	Recall	F1 Score
93.49%	93.52%	93.46%

compared to other sign languages. As shown in Table II, three performance evaluation metrics, such as precision, recall, and F1 score, are calculated from the confusion matrix. RF-Sign system performance is good, with all of its evaluation indexes above 92.8%.

D. Robustness to Various Directional Deviations

RF-Sign performs quite well in terms of recognition accuracy, with an average accuracy of 91% for small deviation patterns and 89% for large deviation patterns. First, perform 36 finger micromovements and measure the horizontal rotation angle of the hand. As shown in Fig. 16, RF-Sign exhibits an average angle error of 4.3° . The variability in error is primarily due to the specific characteristics of certain sign languages, such as finger crossing, which interferes with signal transmission and leads to delayed tag responses. Next, we evaluate the effect of the directional deviation of micromovements on the accuracy performance. We set the hand orientation deviation as no deviation (0°), slight deviation (5° – 10°), and significant deviation (20° – 30°). Fig. 17(a) plots the recognition accuracy under different directional deviations.

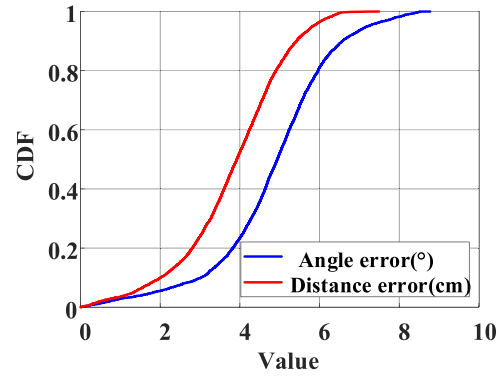


Fig. 16. Angle and distance errors of finger micromovements.

We observe that the accuracy decreases in the case of slight and significant deviations, but the fluctuations are not obvious. Because we construct a physical model that can estimate the angle reasonably and reconstruct the phase profiles so that the original phase changes are maintained. The trend of the phase profiles needs to be consistent, and if deviations occur, they significantly reduce their recognition accuracy.

E. Influence of Distance Between the User and the Antenna

RF-Sign achieves good performance in terms of recognition accuracy when the distance varies, with an average accuracy of 90% for RF-Sign. First, perform 36 finger micromovements and measure the distance from the hand to the antenna. As shown in Fig. 16, RF-Sign exhibits an average distance error of 3.24 cm. Next, we evaluate the activity range of the tag from 0.6 to 1.5 m from the antenna and discuss the effect of the position of every 30 cm within the activity range on the recognition accuracy. Fig. 17(b) plots the recognition accuracy at different distances. We find that the accuracy tends to decrease with increasing distance, specifically because the backscatter communication becomes weak at too great a distance. However, our models use average taking in radial distance estimation to reduce the error and are able to reconstruct the phase profiles to maintain the phase variation trend. As a result, the accuracy remains stable at more than 85%.

F. Robustness to Various Movement Speeds

RF-Sign performs quite well in terms of recognition accuracy, with an average accuracy of 89% in high-speed mode, 92% in medium-speed mode, and 95% in low-speed mode. We set the duration of finger micromovements to three different levels, i.e., high speed (0.5–0.7 s), medium speed (1–1.5 s), and low speed (2–3 s). Fig. 18 plots the recognition accuracy at different moving speeds. Most of the extracted phase profiles are at low to medium speeds. Since the sampling rate is the same, the speed of hand movement is inversely proportional to the length of the phase curve. Therefore, we use resampling to compensate for the effect of hand velocity on the phase before training the model in order to improve the accuracy of the model. While accuracy decreases slightly at high speeds, the performance degradation is mainly due to the tag's inability to respond promptly.

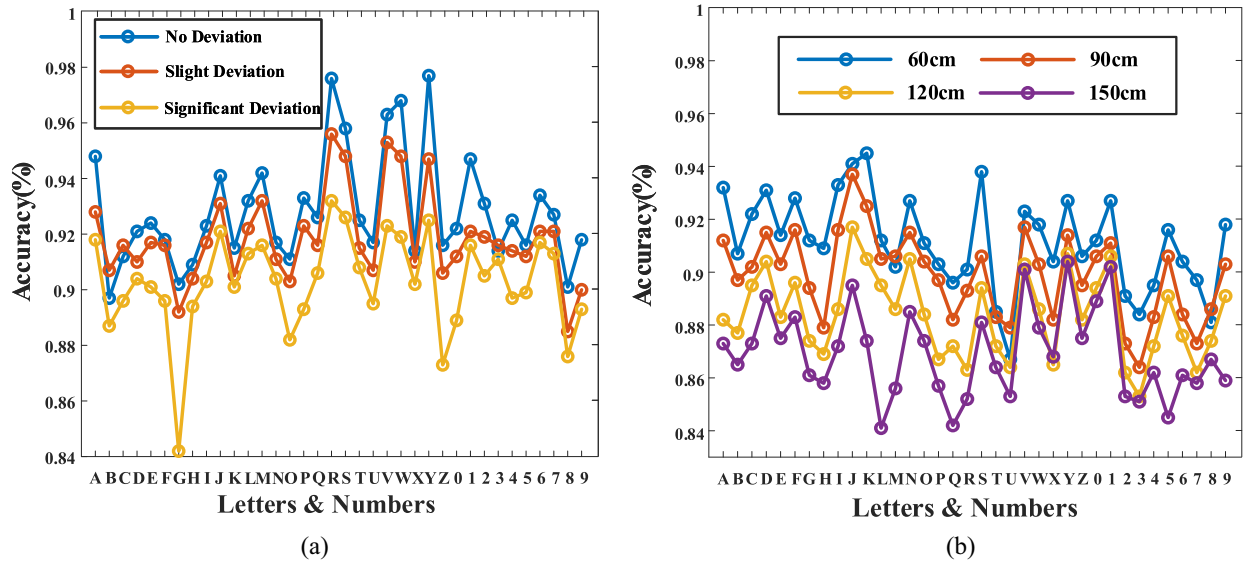


Fig. 17. Recognition accuracy with various directional deviations and various distances. (a) Recognition accuracy of various directional deviations. (b) Recognition accuracy of various distances.

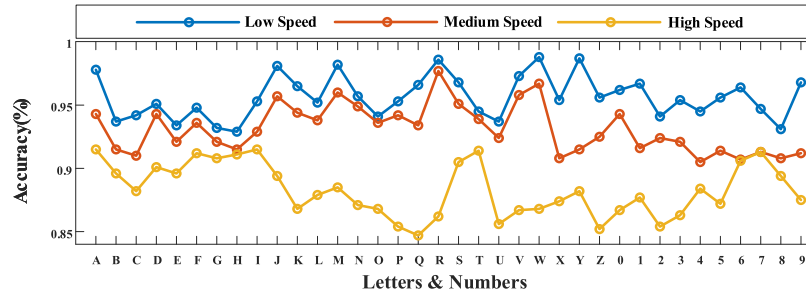


Fig. 18. Recognition accuracy for different movement speeds.

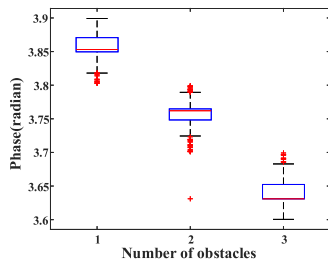


Fig. 19. Phase error of different NLOS environments.

G. Obstacles in the NLOS Environment

RF-Sign can still accurately track gestures in the non-line-of-sight (NLOS) environment. In practical application scenarios, there may be different numbers of obstacles that block the line-of-sight (LOS) path between the user and the RFID antenna. Therefore, it is essential to study the performance of RF-Sign in these NLOS environments. In particular, we investigate the variation of phase measurements when different numbers of obstacles block the LOS path. The number of obstacles is set to 1–3 and placed parallel. As shown in Fig. 19, we find that the phase decreases when the number of obstacles increases. This means that RF-Sign can work in something like free space, even if obstacles block

the LOS path, but vision-based methods cannot work in this case.

Different sizes of obstacles obstruct the LOS path between the hand and the antenna. First, we investigate the phase measurement variations when the LOS path is obstructed by obstacles of different sizes. In the experimental setup, the distance between the antennas is 100 cm. To ensure the normal response of the tags in most cases, the dimensions of the obstacles should be smaller than 100 cm. Square cardboard sheets are used as obstacles with side lengths ranging from 20, 50, to 80 cm. As shown in Fig. 20(a), the phase measurement decreases gradually as the side length range increases, but the difference is not significant. Next, we discuss the impact of obstacle size on recognition accuracy. Fig. 20(b) presents the recognition accuracy for different obstacle sizes. It is found that the accuracy tends to degrade with increasing obstacle edge length, mainly because backscatter communications cause missing values under NLOS paths. However, the RF-Sign system utilizes signal interpolation to reduce errors and maintain the trend of phase variations. As a result, the accuracy remains stable at over 88%.

Different types of obstacles obstruct the LOS path between the hand and the antenna. First, we investigate the phase measurement variations when different types of obstacles (as

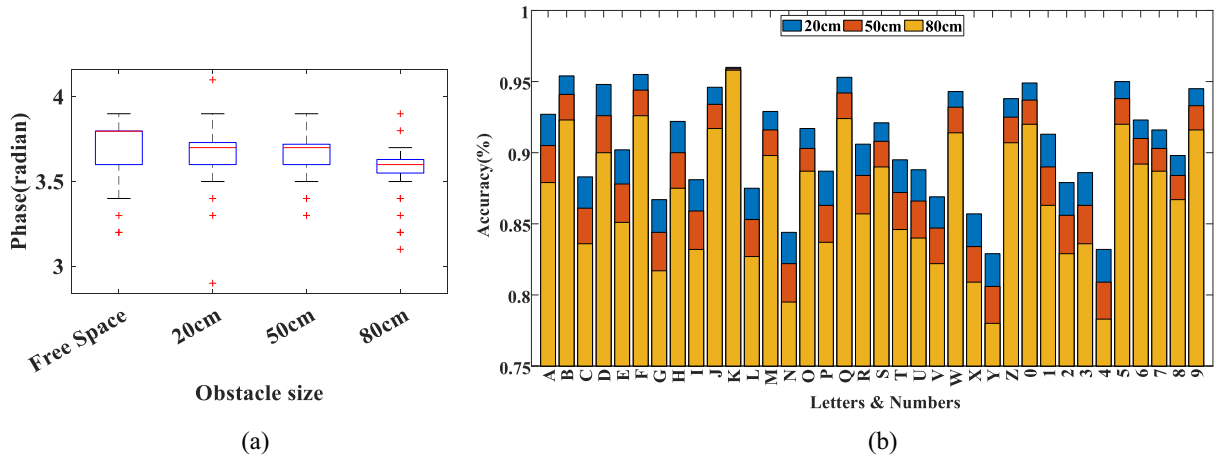


Fig. 20. Phase measurement and recognition accuracy for different obstacle sizes. (a) Phase measurement of different-sized obstacles. (b) Recognition accuracy of different-sized obstacles.

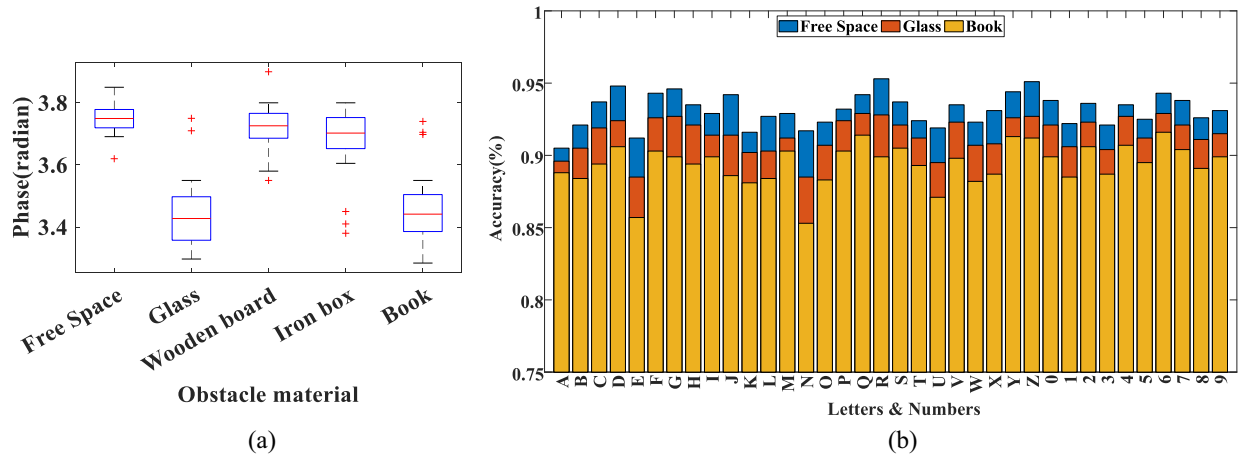


Fig. 21. Phase measurement and recognition accuracy for different obstacle materials. (a) Phase measurements for obstacles of different materials. (b) Recognition accuracy for obstacles of different materials.

shown in Fig. 22, there are wooden board, glass, book, and iron box) are placed along the LOS path. As shown in Fig. 21(a), the phase measurements for books and glass differ by less than 0.53 radians compared to free space, while the phase values for wooden boards and iron boxes remain almost unchanged. Next, we evaluate the recognition accuracy of the RF-Sign system when books and glass obstruct the LOS path. As depicted in Fig. 21(b), it can be observed that due to the impact of multipath effects, the recognition accuracy for these two types of obstacles is slightly lower than in free space. However, the RF-Sign system performs well in NLOS paths, as the multipath effects caused by obstacles are mitigated through the computation of phase differences from finger micromovements and the use of moving average filters.

H. Recognition Delays

RF-Sign achieves good performance in recognizing latency. The average delay for RF-Sign is 289.7 ms. As shown in Table III, we evaluated finger micromovements recognition delays, including time delays in data transmission, signal pre-processing, phase profiles construction, continuous gesture

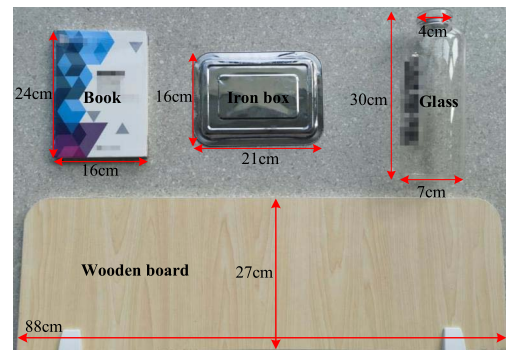


Fig. 22. Different types of obstacles.

segmentation, and sign language recognition, for the different processing steps mentioned above. Relatively high time delays are observed for phase profile construction and continuous gesture segmentation because the length of everyday communication utterances determines the time of the segmentation algorithms, and estimating radial distances, angles, and relative horizontal distances creates time-consuming problems. However, the average latency of each step is less than 73.7 ms.

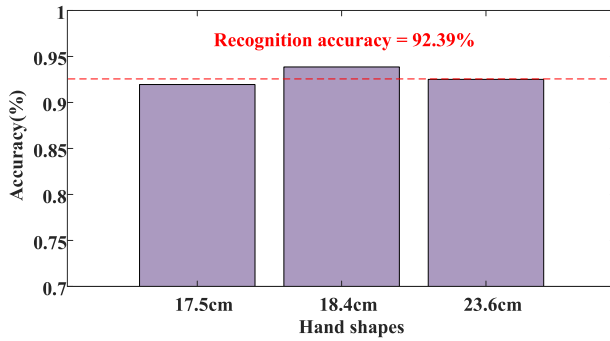


Fig. 23. Recognition accuracy for different hand sizes.

TABLE III
TIME DELAY FOR DIFFERENT PROCESSING STEPS

Processing Step	Time Delay
Data Transmission	52.9 ms
Signal Preprocessing	73.7 ms
Phase Profiles Construction	62.1 ms
Continuous Gesture Segmentation	55.2 ms
Sign Language Recognition	45.8 ms

This is quite low for conventional applications, since micro-movements of the fingers usually last 1–3 s.

I. Robustness to User Hand Shapes

In terms of different hand shapes, RF-Sign performs remarkably well in recognition accuracy, with an average accuracy of over 91.33%. We consider five different hand sizes, ranging from 14.3 to 24.8 cm in wrist-to-middle-finger distance. Among them, the hand sizes of females (first two sizes) are generally smaller and more slender compared to males. We use the sizes of these five hands as our training data set to train the model. Then, we collect a new data set to serve as the test set, which includes three hand sizes: 17.5, 18.4, and 23.6 cm. As shown in Fig. 23, the results indicate that different hand shapes have minimal impact on the recognition accuracy. This is because the communication between RFID tags and antennas is not affected by the size of the fingers, as the size of RFID tags remains unchanged.

J. Effectiveness Against Different Environmental Conditions

In various environmental conditions, as shown in Fig. 23, we consider four scenarios. The data set collected at Location 1 is used to train the model, while the other locations are used to test the model's cross-environment performance. We recollect data sets at Locations 2–4 to assess the model's robustness. As depicted in Fig. 24, RF-Sign exhibits the lowest accuracy at Location 4. This is due to the stronger multipath reflections from nearby objects such as tables, chairs, and walls. However, RF-Sign effectively calculates the tag's phase variations and mitigates the multipath effects by using a moving average filter to smooth the phase profile. As a result,

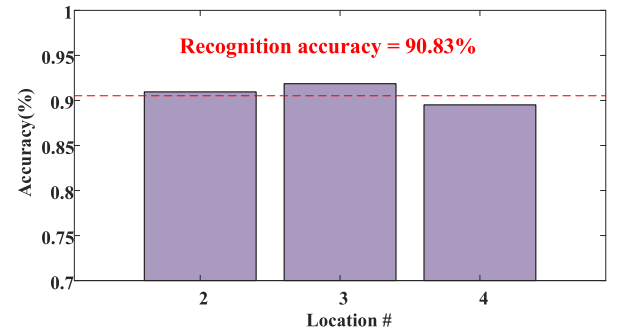


Fig. 24. Impact of different locations on recognition accuracy.

the accuracy at the three locations remains relatively consistent. These results demonstrate the effectiveness of RF-Sign in resisting environmental interference.

V. LIMITATION AND DISCUSSION

In this section, we discuss practical issues related to the application of RF-Sign.

Distance From Antenna Plane: RF-Sign excels in capturing fine-grained micromovements similar to a glove device. However, it has a soft limitation on the distance between the user's hand and the antenna plane. When this distance is within 150 cm, the RF-Sign system demonstrates excellent performance. Beyond this range, the accuracy decreases as the distance increases. This limitation arises because passive tags operate without an external power source and rely on the energy from the RF waves emitted by the antenna. When users move farther away from the antenna plane, the internal antennas of the tags may not receive sufficient power for precise and detailed hand gesture perception. To address this, our future work includes advancing RF-Sign's hardware and signal processing algorithms to overcome this constraint.

Ability to Segment Continuous Gestures: The segmentation algorithm of RF-Sign exhibits varying accuracy for different sign languages. Specifically, we are conducting 20 random combinations of letters or numbers from the misclassification groups (as shown in Table I in Section IV-C) as they represent multiple sign language classes. To better describe the ability to segment continuous gestures, we are calculating the accuracy based on the five samples before and after the correct segmentation point as an error. As shown in Table IV, the average segmentation accuracy for these ten groups is relatively consistent, but it does not reach the level of precise segmentation. This is because fast hand movements may result in insufficient sampling, causing misalignment between the segmented RSS and phase. Thus, one of our future works is to optimize RF-Sign's system parameters, such as reducing the communication time for each tag, to alleviate this issue.

User Convenience: For the current users, wearing a glove-like device to perform sign language communication may cause frustration or reluctance. The reason is that glove devices may not fit all hand sizes, leading to discomfort during prolonged usage. Therefore, we are exploring a potential direction to address this issue in the future, where sign language recognition does not require wearing any device.

TABLE IV
AVERAGE SEGMENTATION ACCURACY OF RF-SIGN

Misgroups	Average Segmentation Accuracy
M1	96.33%
M2	98.19%
M3	94.57%
M4	97.25%
M5	98.29%
M6	95.36%
M7	93.96%
M8	93.11%
M9	96.52%
M10	97.01%

VI. RELATED WORK

Existing research on gesture recognition can be divided into three main categories: CV-based, motion sensor-based, and wireless signal-based approaches.

CV-Based: The image captured by the recording device enables static gesture recognition, while the video enables dynamic gesture recognition. Therefore, hand gesture recognition has a great development in the direction of CV. Previous works have studied static gestures [1], [26], Störing et al. [26] proposed a lookup table-based color segmentation and fast gesture recognition method, but the user lacks interactive experience. Currently, more attention is paid to dynamic gesture recognition in other works [2], [27]. Kýlyboz and Güdükbay [2] used a position tracker containing six degrees of freedom to collect trajectory data and represent the gestures as a 2-D sequence of ordered directional movements. These efforts have common problems: they tend to compromise user privacy and greatly reduce the accuracy of gesture recognition in lightless conditions. RF-Sign relies on RF equipment that can overcome these difficulties and maintain high signal quality in both LOS and NLOS channel environments. In addition, it also shows good performance in coarse and fine particle size.

Motion Sensor-Based: The multimodal characteristics of the accelerometer and gyroscope readings and EMG signals [7], [8], [9], [10], [11] built into the smartwatch allow for good detection of hand motion. For example, Kim et al. [28] proposed an efficient hand gesture recognition (HGR) algorithm that processes time-dependent data from inertial measurement unit (IMU) sensors. However, most of the sensors of wearable devices can determine coarse-grained motion, so they are not effective for fine-grained motion detection. In addition, special sensor devices developed by individuals can recognize some of the sign languages and letters and consider certain problems in terms of reliability and usability, i.e., the limitation of battery life and the possible poor user experience [29]. Considering the above system issues, RF-Sign aims to track fine-grained hand movements by applying RFID tags to individual fingers. In addition, the price of the tag is very low as well as the passive induction type, so there is no need to consider its limited lifetime.

Wireless Signal-Based: Since wireless signals are prevalent in everyday life, gesture recognition has been increasingly studied. For example, radar, WiFi, and acoustic technologies are heavily investigated for gesture recognition [12], [13], [14], [18], [30], [31], [32]. However, these solutions either struggle to perceive fine-grained gestures or fail to resist environmental noise interference. Wifinger [18] perceives and identifies subtle movements of finger gestures by examining the unique patterns exhibited in detailed channel state information (CSI). Recently, RFID-based hand tracking studies [15], [19], [21], [22], [23], [24], [33], [33], [34], [35] are increasing, so sensing hand motion has become a major trend. GRfid [19] proposed that RFID phase information can capture the spatial characteristics of various gestures with low-cost commodity hardware. Zhao et al. [36] extracted gesture-discriminative, user-independent, environment-independent, and deployment-independent features from RFID signals using carefully designed signal processing and adversarial learning. However, these methods do not take into account the effect of rotation angle on the RFID signal, as well as the inability to capture fine-grained finger micromovements. Xie et al. [16] made 3-D modeling of large-range and small-range movements to improve the matching resolution, but arrange a more significant number of antennas. RF-Sign simulates the variation of tag rotation angle and tag distance to dual antennas on phase by constructing a position model, and designing novel methods to cut continuous gestures, so our method is more robust and challenging.

VII. CONCLUSION

In this article, we present RF-Sign, a system that uses RF signals to capture multiple finger micromovements simultaneously. It uses USRP and passive tags. We propose a position-independent finger micromovements recognition method. The implementation of RF-Sign was executed utilizing a USRP system. Subsequent performance evaluations were carried out in a real-world setting, yielding an average accuracy rate of 92.81% across 36 different finger micromovement types.

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